HajjGuard Project Features

Simple project summary for the supervisor

HajjGuard is one responsive web application for pilgrims and authorized operations staff. It helps pilgrims understand crowd conditions, stay connected to their group, and request help. It helps staff view zones, respond to safety incidents, and review system activity.

What the doctor's plan includes

- Pilgrim registration, login, profile, language, age band, and mobility information.
- Joining a Hajj group with a code and viewing the group members and leader contact.
- A map with colored zone status: comfortable, busy, or crowded.
- Current crowd information and a simple forecast for the next 20 minutes.
- A risk score that explains crowd, vulnerability, and heat-related factors.
- Routes that can suggest another corridor when a zone is becoming crowded.
- A large SOS action with category and location information.
- QR-based lost-pilgrim recovery and the group leader contact.
- Operations screens for zones, alerts, SOS tickets, reports, and user management.
- Arabic and English responsive screens for phone and laptop use.

Extra features already made in HajjGuard

- Real geographic map tiles with roads and places around the Holy Sites; no decorative fake map is used.
- Privacy and account protection: authenticated access, safer ownership checks, explicit consent, and protected sensitive actions.
- Consent-based location sharing that can be revoked and expires with the pilgrim’s trip.
- Group safety tools: drift and geofence alerts, group messages, phone ping requests, and notification support.
- Offline emergency card with saved trip details, group contact, emergency numbers, and QR card.
- Secure QR cards using random revocable tokens instead of predictable personal identifiers.
- SOS emergency routing by category: medical or heat 997, rescue or fire 998, and lost, security, or general 911.
- A user-confirmed call handoff: the phone opens its call screen, while HajjGuard records that it cannot verify the call was answered.
- Safety map layers show only source-labelled verified facilities; unverified points are not invented.
- Clear labels for synthetic, local, or unavailable data so a demonstration is not presented as live field data.

What the app will do in a simple user journey

- A pilgrim signs in, joins a group, and sees the map and current zone status.
- The pilgrim can choose to share location for the trip and can stop sharing at any time.
- The pilgrim receives a warning if they drift from the group or approach a configured safety boundary.
- The pilgrim can open the offline card, send an SOS, or manually call the selected emergency number.
- Staff see authorized zones and incidents, review alerts, and maintain records through the operations screens.

Important project boundaries

- The project does not claim official Ministry, Nusuk, Nafath, Civil Defense, SCADA, CCTV, wallet, or emergency-service integration.
- It does not place emergency calls automatically. It provides a manual call link after user confirmation.
- Crowd, prediction, and route demonstrations use local or synthetic data unless a source is clearly shown.
- The system is an academic safety and decision-support prototype. Field validation and official approval are future work.

Final message for the presentation

HajjGuard brings crowd awareness, group safety, lost-pilgrim recovery, and emergency support into one simple web application. The project follows the doctor’s required baseline and adds stronger privacy, consent, offline access, secure QR recovery, group alerts, and a usable geographic map.